A: Solving the differential equations

According to the assumption in (3.1) the model point moves along the s-axis, subject to an acceleration $a = \frac{dv}{dt}$ given by

$$\frac{dv}{dt} = -\alpha v - \beta v^2, \qquad \text{(A1)}$$

where α and β are constants. Proving the result (3.2) amounts to showing that if the model point starts at $s = 0$ with $v = v_0$, its velocity at the point s on the circular s-axis must be given by

$$v(s) = (v_0 + w)\,e^{-s\beta} - w \;\; ; \; w = \frac{\alpha}{\beta}\,, \qquad \text{(A2)}$$

To show this, we first insert the rearrangement

$$\frac{dv(s)}{dt} = \frac{dv(s)}{ds}\frac{ds}{dt} = \frac{dv}{ds}v\,, \qquad \text{(A3)}$$

in (A1), multiply by $\frac{\beta}{v}$ on both sides and get a result which can be written as

$$\frac{d(\alpha+\beta v)}{ds} = -\beta(\alpha+\beta v)\,, \qquad \text{(A4)}$$

(With $\frac{d\alpha}{ds} = 0$, the left side equals $\beta\frac{dv}{ds}$). In the complete solution to (A4),

$$\alpha+\beta v = C\,\exp(-\beta s) \qquad \text{(A5)}$$

the constant C must be chosen so as to get $v = v_0$ ($\Leftrightarrow \alpha+\beta v = \alpha+\beta v_0$) when $s = 0$. Hence we must choose $C = \alpha+\beta v_0$ which gives

$$\alpha+\beta v = (\alpha+\beta v_0)\exp(-\beta s) \qquad \text{(A6)}$$

From this, the result aimed at follows directly by isolating v.

A paradox ?

In the limit $s \to \infty$, the velocity v in (A2) approaches the limit $-w$. Clearly something is wrong here. To find out what, we now solve eqn. (A1) so as to obtain the position $s(t)$ as well as the velocity $v(t)$ as functions of the time t. This can be done simply by noting it is an example of the so-called logistic equation[1],

[1] Alternatively it can be solved directly as follows: Divide on both sides with v^2 and rearrange,

$$-v^{-2}\frac{dv}{dt} = \alpha v^{-1} + \beta \;\Leftrightarrow\; \frac{dv^{-1}}{dt} = \alpha v^{-1} + \beta \;\Leftrightarrow\; \frac{d(v^{-1}+\frac{\beta}{\alpha})}{dt} = \alpha(v^{-1} + \frac{\beta}{\alpha})$$

The complete solution to the last equation is

$$(v^{-1} + \tfrac{\beta}{\alpha}) = ke^{\alpha t}$$

where k is an arbitrary constant. Isolating v one finds

$$v = \frac{1}{ke^{\alpha t} - \frac{\beta}{\alpha}} = \frac{\frac{\alpha}{\beta}}{1+c\,e^{\alpha t}}$$

where c is another arbitrary constant.

known to be solved by

$$v(t) = \frac{\frac{-\alpha}{\beta}}{1+c\exp(\alpha t)} \tag{A7}$$

where the constant c is arbitrary. By the initial condition $v(0) = v_0 > 0$ we get $v_0 = \frac{-w}{1+c} \Leftrightarrow 1+c = -\frac{w}{v_0} \Leftrightarrow c = -(1+\frac{w}{v_0})$ and hence

$$v(t) = \frac{-w}{1-(1+\frac{w}{v_0})\exp(\alpha t)} \tag{A8}$$

It is convenient to write this result in 3 other different ways,

$$v(t) = \begin{cases} \dfrac{w}{(1+\frac{w}{v_0})e^{\alpha t}-1} \\ \dfrac{v_0}{(1+\frac{v_0}{w})e^{\alpha t}-\frac{v_0}{w}} \\ \dfrac{v_0 e^{-\alpha t}}{1+\frac{v_0}{w}(1-e^{-\alpha t})} \end{cases} \quad \text{where } w = \frac{\alpha}{\beta} \quad . \tag{A9}$$

The second expression is obtained by multiplying the first in nominator and denominator by $\frac{v_0}{w}$ and the last expression is obtained by multiplying once more with $e^{-\alpha t}$. Next, integration of $v(t)$ gives[2]

$$s(t) = \frac{1}{\beta}\ln(1+\frac{v_0}{w}(1-e^{-\alpha t})\,) \tag{A10}$$

It is seen that if $t \to \infty$, then s approaches the limit

$$s_\infty = \frac{1}{\beta}\ln(1+\frac{v_0}{w}) \tag{A11}$$

This resolves the "paradox" mentioned above: The movement slows down so quickly, that s never passes the limit s_∞ .

We note in passing that the result (A10) can be rewritten as

$$e^{\beta s} = 1+\frac{v_0}{w}(1-e^{-\alpha t}) \quad \Leftrightarrow \begin{cases} \frac{v_0}{w}(1-e^{-\alpha t}) = e^{\beta s}-1 \\ e^{-\alpha t} = 1-\frac{w}{v_0}(e^{\beta s}-1) \end{cases} . \tag{A12}$$

Inserting the last of these results in the last result for $v(t)$ in (A9) we get

$$v = \frac{v_0(1-\frac{w}{v_0}(e^{\beta s}-1))}{1+e^{\beta s}-1} = \frac{v_0-w(e^{\beta s}-1)}{e^{\beta s}} = (v_0+w)e^{-\beta s}-w \tag{A13}$$

This verifies the result obtained in another way in (A6).

An interesting expression

Isolating of t in the lower expression in (A12) one finds

$$t(s) = -\frac{1}{\alpha}\ln(1-\frac{w}{v_0}(e^{\beta s}-1)) \tag{A14}$$

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[2] Check: $\frac{ds}{dt} = \frac{1}{\beta}\frac{1}{1+\frac{v_0}{w}(1-e^{-\alpha t})}\cdot\frac{v_0}{w}(-e^{-\alpha t})(-\alpha) = \frac{v_0 e^{-\alpha t}}{1+\frac{v_0}{w}(1-e^{-\alpha t})}$ and $s(0) = \frac{1}{\beta}\ln(1) = 0$

This suggests an alternative way of using the merry-go-round. It means that the model point on the circular s-axis should pass the positions $s_n = 2\pi R \cdot n$; $n = 0, 1, 2, \ldots$ at the times $0, t_1, t_2, \ldots.$ given by

$$t_n = -\frac{1}{\alpha} \ln(1 - \frac{w}{v_0}(e^{2\pi R\beta \cdot n} - 1)) \qquad \text{(A15)}$$

These times t_n for the passages could be registered by the photo gate and (v_0, α, β) could be determined by fitting to the expression in (A15). Thereby we would no longer have to worry about the accuracy of the various lengths other than R needed for the measurements of the decreasing velocities $u_0, u_1, u_2, \ldots$. And the times needed for these cannot be as accurately determined as the much longer times in (A15). This possibility has to be investigated further.